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Peatland drainage alters soil structure and water retention properties: Implications for ecosystem function and management

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Abstract

Peatland functions (e.g., carbon sequestration and flora diversity) are largely driven by soil moisture dynamics and thus dependent on interactions between hydrologic regimes and organic soil properties. Understanding these interactions is particularly important in drained peatlands, where drier conditions may alter soil properties with feedbacks to soil water retention and associated ecosystem functions. In this work, we focused on the Great Dismal Swamp (GDS) in Virginia, USA, a historically drained, temperate peatland with ongoing hydrologic restoration efforts. Two distinct soil layers varying in thickness exist at GDS: an upper layer with subangular blocky structure thought to be a result of past drainage, and a sapric lower layer with a massive structure more representative of an undisturbed state. To understand the occurrence and consequences of these distinct layers, we used continuous water table data and analysed soil physical and hydraulic properties to characterize soil profiles at 16 locations. We found significant differences between layer properties, where upper layers had lower fibre and organic matter contents and higher bulk densities. Further, moisture release curves demonstrated lower water retention in upper layers compared with lower layers and key differences in pore structure, with upper layers having higher macroporosity. Upper layers varied in thickness across sampling locations (~0.30 to 1.0 m) with a transition to lower soil layers typically occurring at depths below contemporary water level observations, suggesting that the upper layer may be a result of historical drainage and deeper water table conditions. Yet, upper layers with more frequent saturation exhibited higher water retention and lower macroporosity compared with drier upper layers, thus indicating potential recovery following re-wetting efforts. These findings highlight how past drainage influences soil properties and water retention, with important implications for current management objectives at GDS and other drained peatland systems.

KEYWORDS
disturbance, great dismal swamp, hydrologic restoration, moisture release curves, peatland, soil properties, soil shrinkage, water management

1 | INTRODUCTION

Peatlands are the most prevalent wetland type in the world and are characterized by deep (>30–40 cm) organic soils with more than 30% organic matter (Joosten & Clarke, 2002; Rydin et al., 2013; Soil Survey Staff, 2014). Peatlands exist across large latitudinal gradients, with ~80% of the world's peatlands located in boreal regions, 15%–20% located in the tropics and less than 5% in temperate regions (Lappalainen, 1996; Rein et al., 2008). Collectively, they occupy 4 million km^2, ~3% of the Earth's land surface (Xu et al., 2018), but disproportionately contribute to terrestrial ecosystem services, including carbon sequestration, water quality and storage, and habitat for unique flora and fauna (Joosten & Clarke, 2002). Such functions are largely dependent on interactions between hydrologic regimes and peat soil properties that help to support persistent soil saturation, low decomposition, and associated peat accumulation (Belyea & Clymo, 2001; Clymo, 1984; Waddington et al., 2015).

The degree of decomposition also influences peat soil properties (e.g., bulk density, fibre content, pore structure) and their depth-dependent variation (Verry et al., 2011; Belyea & Clymo, 2001; Boelter, 1968). Younger organic matter deposits that occur above the water table experience more frequent aerobic conditions, and thus have higher decomposition rates, yet are less decomposed than deeper and older layers (Clymo et al., 1998). Consequently, the degree of peat decomposition and thus bulk density typically increase with depth (Boelter, 1968; Clymo, 1984), whereas total porosity, macroporosity and fibre content decrease with depth (Verry et al., 2011; Boelter, 1968). These gradients in physical properties often cause concurrent gradients in important hydraulic properties, such as decreasing hydraulic conductivity and increasing soil water retention with depth (Verry et al., 2011; Boelter, 1968).

Peatlands are commonly degraded via drainage, a widespread human disturbance occurring across ~10% of global peatland area (Leifeld & Menichetti, 2018) with consequences for peat soil properties (Holden et al., 2006). More frequent aerobic conditions from drainage can lead to increased peat soil oxidation, resulting in increased bulk density, decreased fibre content and decreased pore volume (Kechavarzi et al., 2010; Leifeld et al., 2011). Drainage has also been shown to cause peat shrinkage and distinct changes in pore structure and associated moisture holding capacity in upper peat layers (McCarter & Price, 2015; Peng et al., 2007; Peng & Horn, 2007; Schwärzel et al., 2002). Consequently, drainage can result in persistent changes in soil hydraulic properties, which in turn may affect soil moisture regimes. Soil moisture regime drives decomposition rates (and thus carbon sequestration) and vegetation composition, as well as the principal determinant of vulnerability to peat smouldering fires (Frandsen, 1997). As such, changes in soil hydraulic properties may represent an additional consequence to ecosystem function that could persist following rewetting efforts in drained systems.

Following drainage, attempts to restore peatland hydrology often focus on installing water control structures in ditches or via ditch filling (Chimner et al., 2016). However, changes in water retention properties of degraded peat soils present an important, and less recognized, challenge for such restoration efforts. Quantifying the extent and magnitude of changes to peat physical and hydraulic properties following drainage is necessary to guide restoration of drained peatland ecosystems. While there has been recent work characterizing degraded peat properties in northern bog and fen systems (e.g., Gauthier et al., 2018; McCarter & Price, 2015; Taylor & Price, 2015), limited research has been conducted in other peatland systems, particularly those with differing organic soil types (e.g., sapric vs. fibric). To that end, here we focus on the Great Dismal Swamp (GDS) National Wildlife Refuge (hereafter GDS), an intensively drained, temperate forested peatland largely dominated by sapric soils.

GDS protects a 45 000-ha remnant of a forested peatland that once extended over more than 500 000 ha (Osborn, 1919) of southeastern Virginia and northeastern North Carolina (Figure 1). Historically, GDS was characterized by peat depths exceeding 4 m and forested stands of Atlantic white-cedar (*Chamaecyparis thyoides*), bald-cypress (*Taxodium distichum*), and swamp tupelo (*Nyssa biflora*) (Levy, 1991; Whitehead, 1972). Beginning in the mid-18th century, drainage ditches were constructed throughout GDS to lower the water table and facilitate timber harvesting (Levy, 1991). Drier conditions resulted in vegetation shifts to communities now dominated by red maple (*Acer rubrum*) (Atkinson et al., 2003), soil oxidation and subsidence and increased vulnerability to soil-consuming, smouldering fires (Wurster et al., 2016). Indeed, two recent wildfires in GDS burned up to 1 m of peat and released ~1.7 Tg of carbon to the atmosphere (Sleeter et al., 2017). In response, water control structures have been installed throughout the drainage ditch network to raise the water table and restore historical soil moisture regimes (Wurster et al., 2016).

Drainage at GDS also likely altered soil properties as evidenced by two distinct layers, with the upper layer thought to be a result of historically lower water table conditions (Soil Survey Staff, 2020). Such persistent effects of past drainage have potentially important consequences for contemporary water retention properties and resulting soil moisture regimes. Given the limited work on soil hydraulic properties in drained temperate peatlands, here we addressed two objectives: (i) compare the physical and hydraulic properties of upper and lower layers and (ii) assess the influence of contemporary water levels on upper layer properties and thus their potential recovery following re-wetting efforts. In addressing these objectives, our goal was to inform the ongoing restoration efforts at GDS and other drained peatlands more broadly.

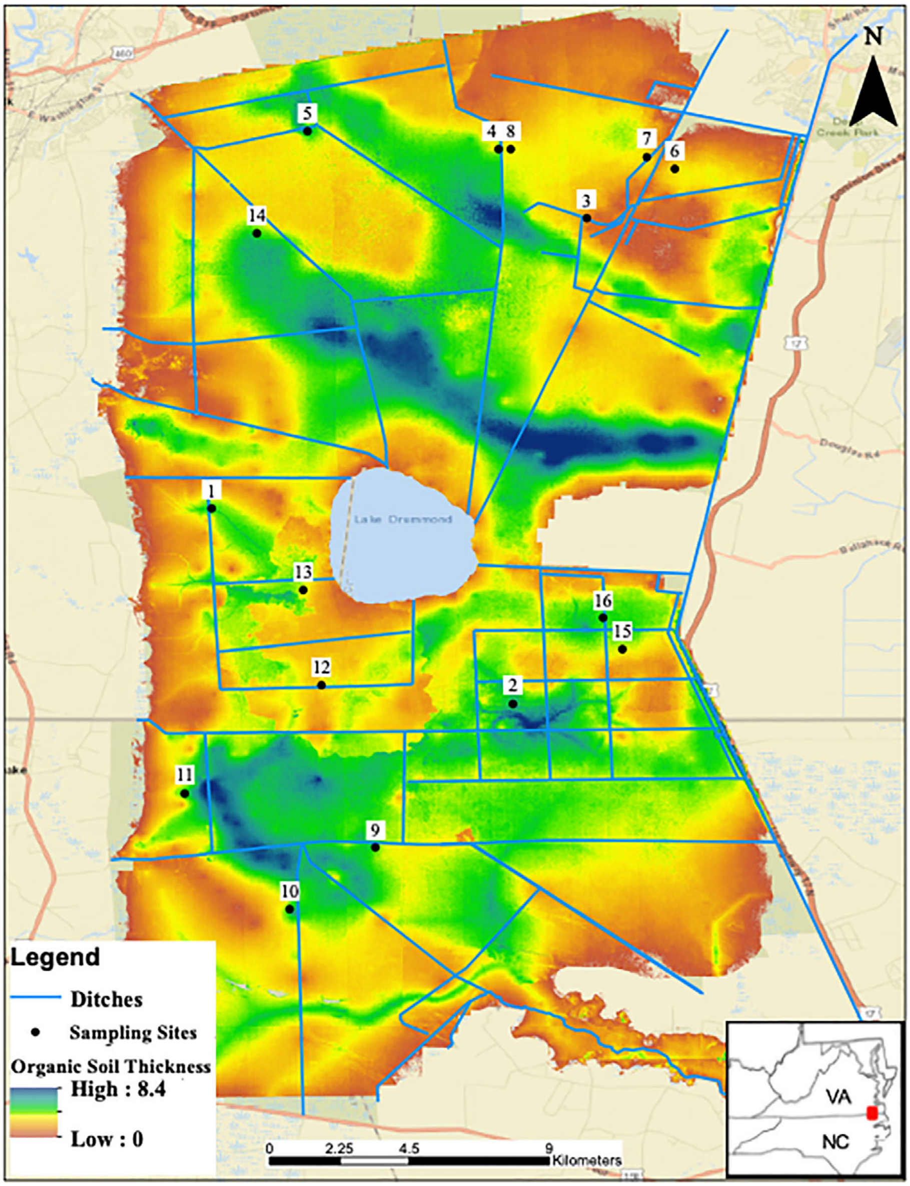

FIGURE 1 Map of soil thickness, site locations, and ditch networks at the Great Dismal Swamp National Wildlife Refuge

2 | METHODS

2.1 | Study system and site selection

GDS protects 45 000 ha of forested peatlands on the border of southeastern Virginia and northeastern North Carolina (36.6201°, −76.5599°). Climate at GDS is temperate, characterized by long humid summers and mild winters. Mean annual precipitation is 1280 mm, and mean annual temperature is 16°C (1981–2018; PRISM Climate Group, 2019). Peat accumulation began ca. 9000 years ago due to low topographic relief and a clay-rich confining layer, with historical peat depths ranging 1–5 m (Whitehead, 1972) Marsh species, such as grasses and sedges, are thought to be the initial botanical source for peat accumulation, later followed by woody vegetation as forested communities developed (Ingram & Otte, 1981; Lewis & Cocke, 1929). Contemporary organic soils at GDS remain deep, although now with a shallower range due to oxidation (0.3 to 4 m; Reddy et al., 2015) and are generally mapped as Typic Haplosaprists (Soil Survey Staff, 2020). Further, two distinct organic soil layers now occur at GDS: an upper layer that has a weak subangular blocky structure and thought to be a result of drainage, and a lower layer described as sapric with massive structure (Soil Survey Staff, 2020).

To evaluate spatial variation in soil layer thicknesses, water table levels, and soil physical and hydraulic properties, we established 16 sampling sites co-located with existing water table monitoring wells (managed by GDS staff) (Figure 1). Sites were selected under the following criteria: they existed on organic soil deposits; had overlapping water table data of at least 2.5 years; and had consistent water level management over that time period (i.e., no changes to proximate water control structures). Each site was centred on a well and consisted of three plots located 8 m radially from the well, on azimuths of 0°, 120° and 240°. If a plot location fell on any obstruction (e.g., tree, cypress knees, roots), the transects were rotated 10° clockwise until there were no obstructions.

2.2 | Soil profile and water table characterization

At each plot, we measured total peat soil depth and upper- and lower-layer thicknesses. Total peat soil depth was measured using a soil rod, pushed by hand until refusal by the underlying mineral layer. We measured individual layer depths and thicknesses using intact cores from a Russian Sediment/Peat Borer (AMS, Inc., American Falls, Idaho). The Russian peat borer was used to take 50 cm long intact cores, extracted sequentially until all organic layers were represented. To reduce disturbance, multiple side-by-side holes were used to obtain samples with the Russian peat borer. We identified layers in the field by qualitative characteristics, where upper layers were characterized by coarse, subangular blocky structure and lower layers by their massively structured muck, more typical of undisturbed Haplosaprists (Soil Survey Staff, 2020). Then we measured each layer for depth and thickness.

Sub-daily (30 min) water table data from March 2017 to September 2019 were obtained at each site from either the USGS National Water Information System (U.S. Geological Survey, 2016) or GDS staff, depending on well ownership. To estimate water table depth regimes at each plot, we surveyed each plot's ground surface elevation relative to the ground surface at the well (i.e., site centre). We then calculated water table depths based on the difference between the surface elevations at each plot and the water elevations at the centre of the site, assuming a locally flat water table over the 8 m distance between the well and each plot location.

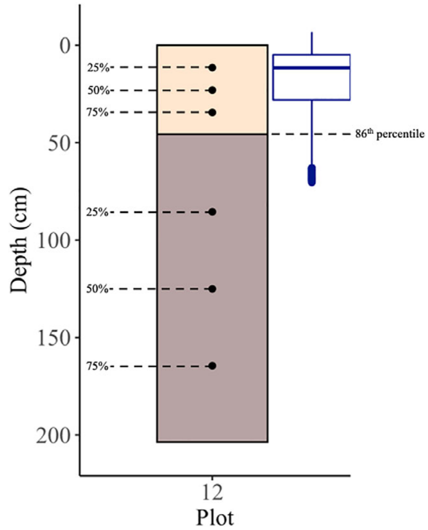

FIGURE 2 An example soil sampling profile and water table relationships using Site 12. Total peat soil depth was 204 cm. Upper layer thickness was 46 cm and bulk density core samples were taken at the 11.5, 23 and 34.5 cm within the soil profile to correspond to the 25th, 50th, and 75th depth percentiles within the layer. The lower layer was 158 cm thick and samples were taken at 85.5, 125 and 164.5 cm within the profile to correspond to the 25th, 50th and 75th depth percentiles. The water table was calculated to cross the bottom boundary of the upper layer in the 86th percentile

2.3 | Soil sampling and analysis

At one plot per site (the 0° azimuth location), we obtained soil samples from the 25th, 50th and 75th depth percentiles of each observed layer based on characterized layer thicknesses (see example in Figure 2). We chose percentiles as opposed to consistent depths because upper and lower layers exhibited large variation in thicknesses across sites, and our goal was to characterize within-layer variation using distributed samples along their full extent. If the observed layer was less than 40 cm thick, then only one sample at the layer midpoint (50th percentile) was collected. Laboratory analyses for bulk density and hydraulic properties required intact, field-representative samples. For these samples, we used two separate sampling methods due to differences in upper and lower soil structure and position. For all upper layer samples, a 5 cm diameter × 10 cm long (comprised of two 5 cm segments) Bulk Density Corer (AMS, Inc., American Falls, Idaho) was used. We then retained only the lower 5 cm segment to avoid issues of compaction and disturbance. For lower layers, we used a Hand Core Sediment Sampler (Wildco®, Yulee, Florida) to collect samples in 5 cm diameter × 25 cm long plastic tubes at our desired depth range, and then cut each tube down to a single 5 cm × 5 cm sample within our target sampling percentile and to match the upper layer core dimensions. We also collected an unconsolidated sample at each sampling depth for destructive laboratory analyses (organic matter content, fibre content). All samples were capped at both ends, placed in plastic bags and stored at 4°C until further processing and analysis.

For all collected soil samples, we analysed bulk density, organic matter content and fibre content (hereafter referred to as physical properties). We determined bulk density (ρ_b; g cm^{-3}) using the intact samples after oven drying samples at 105°C for 24 h (ASTM, 2008), following hydraulic property analysis (below). Organic matter and fibre content analyses were conducted using the unconsolidated grab samples. Organic matter content was determined through soil mass loss on ignition at 380°C for 24 h (ASTM, 2020). We determined fibre content following ASTM (ASTM, 2013), where we placed samples first in 5% sodium hexametaphosphate solution for at least 15 h, stirred, and then poured over a 100-mesh sieve while washing with distilled deionized water until the water passing through the sieve ran clear. The sieve was then placed in a shallow tank of 2% HCl solution for at least 10 min and re-washed, after which any large mineral grains and pieces of plant material (e.g., roots or wood) greater than 20 mm were removed. The remaining sample was then washed onto pre-weighed filter paper and placed in a drying oven at 105°C for 24 h. We determined fibre content by dividing the dry mass of the fibres by the estimated dry mass of the unconsolidated sample (now consumed), which was based on the initial field mass of the unconsolidated sample and the field and dry mass of the intact core for the same sampling location (ASTM, 2020).

We also analysed soil samples for properties that influence soil water retention (hereafter referred to as hydraulic properties) by developing moisture release curves, which describe the relationship between volumetric soil water content, θ (cm cm^{-3}), and matric potential, h (expressed here as a positive tension; Klute, 1986). For each intact soil sample, we constructed moisture release curves by applying multiple tensions using tension table approaches (Leamer & Shaw, 1941) for low tensions and pressure plate approaches (ASTM International, 2016; Dane & Hopmans, 2002) for higher tensions. Low tensions (15, 30, 45 and 60 cm) were generated by adjusting the height of the levelling device on the tension table relative to the middle of the soil cores. For each tension, the intact soil core samples (5 cm $\times$ 5 cm) were saturated for 24 h, weighed, placed onto the tension table for 24 h and then re-weighed. For high tensions (330 and 1000 cm), the same intact core samples were saturated for 24 h, weighed and placed on a ceramic plate (rated for 1 bar) within the pressure chamber. After samples reached equilibration (i.e., there was no observed mass change from the sample under the given tension), they were removed from the chamber, weighed and re-saturated for the following pressure application. Following the final pressure, samples were oven-dried at 105°C for 24 h and dry mass (M_d) was determined.

For each tension applied, θ was determined using:

$$\theta(h) = \left(\frac{M_h - M_d}{V_t}\right) \tag{1}$$

where M_h is the mass (g) of the sample at the measured tension and V_t is the total sample volume (cm^3), while assuming a density of water equal to 1 g cm^{-3} to convert units. A variation of the Brooks and Corey (1964) model for water retention was used to develop moisture release curves using θ at each applied tension. The function is reduced to:

$$\theta(h) = (\theta_s - \theta_r) * \left(\frac{h_b}{h}\right)^{\frac{(\eta-2)}{3}} + \theta_r \tag{2}$$

where θ_s is water content at saturation (cm^3 cm^{-3}) and parameters h_b (bubbling pressure; cm), η (pore size index; unitless) and θ_r (residual water content; cm^3 cm^{-3}) were optimized, minimizing the sum of squared residuals.

With the developed moisture release curves, we quantified a suite of hydraulic properties including: total porosity (ϕ; cm^3 cm^{-3}), microporosity (ϕ_{mi}; cm^3 cm^{-3}), macroporosity (ϕ_M; cm^3 cm^{-3}), volumetric water content at 1-bar (θ_{1bar}; cm^3 cm^{-3}), and soil capillary length (λ; cm). Total porosity of each sample was determined as equal to θ_s. Macropores were defined as those which drain under gravity, approximately >95 μm (Brady & Weil, 2010). Using the Young-Laplace equation and assuming cylindrical pores, the $h = 15$ cm tension corresponds to an approximate pore radius of 95 μm; therefore, we calculated macroporosity as $\phi_M = \theta_s - \theta_{15}$, where θ_{15} represents volumetric water content at 15 cm of tension. Microporosity was correspondingly calculated as $\phi_{mi} = \theta_s - \phi_M$, representing pores ≤95 μm in radius (Luxmoore, 1981). θ_{1bar} was determined as the measured water content at $h = 1020$ cm (100 kPa), and represented water retention at our largest applied tension. Last, λ was calculated using parameters h_b and η from the modelled moisture release curves following Stewart and Najm (2018):

$$\lambda = \frac{h_b \eta}{(1-\eta)} \tag{3}$$

As such, λ distinguished samples based on their parameterized moisture release functions, as well as directly indicated sample differences in saturation extent above water table levels (i.e., capillary lengths).

2.4 | Comparing upper- and lower-layer properties

To evaluate differences in soil properties between layers, we conducted independent two-sample t tests ($\alpha = 0.05$) on each observed property between the upper and lower layers. Data were tested for normality using the Shapiro–Wilk test ($\alpha = 0.05$) and for equal variances, followed by two-sample student's t test, Welch's t test, or the nonparametric Wilcoxon sum rank as appropriate.

We also assessed relationships among observed physical and hydraulic properties using two different analyses. First, we used a nonmetric multidimensional scaling (NMDS) in R (R Core Team, 2016), using 'metaMDS' function in the 'vegan' package (Oksanen et al., 2019). Soil property data were input as processed and then scaled to normalize using the Wisconsin double standardization. For model development, we used the Bray–Curtis dissimilarity method with 10 000 model iterations to avoid local minimum and maximums and a stress test to determine goodness-of-fit of transformed data in ordination space. The resulting property vectors were fit to the axes using the 'envfit' function to show each parameter's relationship in ordination space. Second, we explored correlations among all soil properties using a Spearman's rank correlation ($\alpha = 0.05$).

2.5 | Relating upper layer properties to water table regimes

From our 2.5-year daily water table records, we determined several metrics for each plot, including mean, median, standard deviation, and 25th, 75th and 95th percentiles. We then conducted a Spearman's correlation ($\alpha = 0.05$) between observed upper layer thicknesses and these water table depth metrics. For this analysis, we excluded one site (13), where we did not observe an upper layer (presumably due to a recent soil-consuming fire). For each plot, we also used cumulative distribution functions (CDF) of water table depths to identify the water table depth percentile that coincided with the bottom boundary of the upper layer (see example in Figure 2). This approach allowed us to assess if upper layer boundaries occurred at shallow (i.e., small percentile values) versus deep water tables experienced at each plot over the 2.5-year record. We re-ran this CDF analysis for plots at the nine sites with existing water table data for at least 4.75 years (compared to the 2.5-year record) to extend inferences to longer time periods. Finally, we examined correlations between upper layer soil properties from the multiple sampling depth locations (25th, 50th and 75th percentiles) and their percentage of time saturated using a Spearman's rank correlation ($\alpha = 0.05$).

3 | RESULTS

3.1 | Soil profile and water table characterization

Peat soil depths, upper and lower layer thicknesses and water table regimes varied across our 16 sites and among the associated measurement plots (Figure 3). Total peat soil depths ranged from 34 to 283 cm. Eight sites had both upper and lower layers present whereas seven sites had only an upper layer. Site 13 was located in a fire scar and was observed to only contain lower peat layers. Upper- and lower-layer thicknesses were variable across sites, ranging from thick upper layers with thin lower layers to thin upper layers with thick lower layers. Upper layer thickness ranged from 34 to 110 cm (mean = 68.3 cm), and lower layer thickness ranged from 14 to 240 cm (mean = 135.9 cm). There was also variability in layer thicknesses among plots within sites, which typically exceeded plot variation in ground surface elevation (see error bars in Figure 3). Sites also experienced variable water table regimes, ranging from inundated sites with daily water tables existing above the ground surface for much of the year, to dry sites with limited inundation, to sites with more variable water tables. Across all sites, water tables varied from 45.5 cm above the soil surface to 125.7 cm below the soil surface. Notably, water tables only dropped below the bottom boundary of the upper layer at seven of our 16 sites: Sites 2, 3, 6, 9, 10 and 12.

3.2 | Comparing upper- and lower-layer properties

We found significant differences in all of the quantified soil physical and hydraulic properties between the upper and lower layers (Table 1). The upper layer samples had significantly higher bulk densities, lower organic matter concentrations and lower fibre contents compared with the lower layer samples. Developed moisture release curves ($n = 43$) consistently demonstrated lower water retention capacities in upper compared with lower layers, with typically lower water contents in the upper layer across all tensions and more rapid water content declines, particularly at low tension values (Figure 4). Upper layers also exhibited significantly higher total macroporosity and significantly lower values for microporosity, capillary length and θ_{1bar} (Table 1), further demonstrating less water retention in upper compared with lower layers.

NMDS analysis using physical and hydraulic soil properties resulted in a two-dimensional solution with a stress of 0.06 (Figure 5, Table 2). Physical properties were aligned along Axis 2, with increasing bulk density opposing fibre content. Organic matter was also largely associated with Axis 2 and opposed bulk density (Table 2), but is not shown in Figure 5 since the association was not significant. Hydraulic properties were aligned with Axis 1, where increasing water retention properties (i.e., capillary length, θ_{1bar}, and microporosity) oppose macroporosity. The ordination also illustrates grouping of upper and lower layers; however, this grouping reveals larger separation along Axis 1 and thus the differences are more associated with hydraulic properties.

Consistent with NMDS findings, we found a negative correlation between bulk density and fibre content (Table 3). Similarly, there were significant negative correlations between macroporosity and the other hydraulic properties. However, there were no significant correlations between physical and hydraulic properties, indicating that variation in physical properties within and between layers did not explain observed variation in water retention properties.

3.3 | Relating upper layer properties and water table regimes

Upper layer thickness was often greater at drier sites (i.e., deeper water tables), as evidenced by significant and positive correlations between

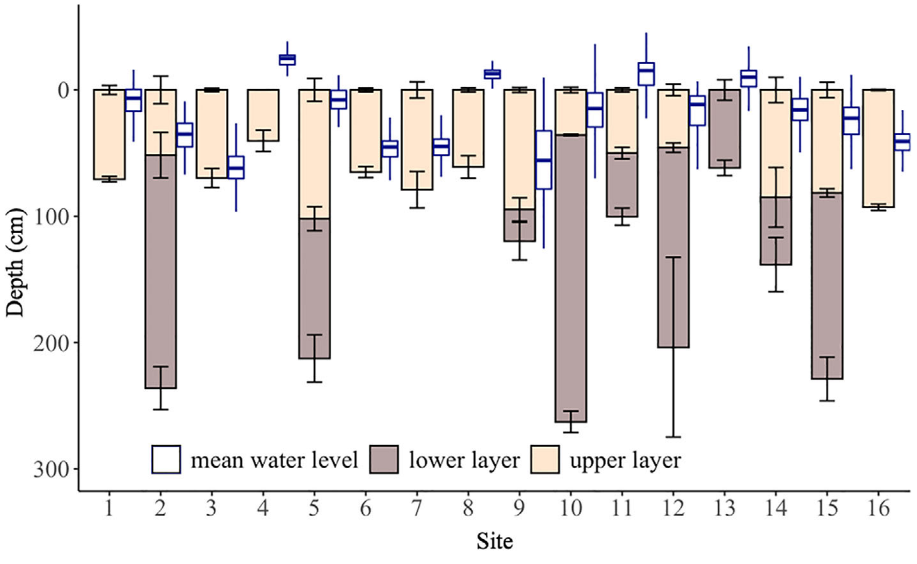

FIGURE 3 Mean relative ground surface elevation, total peat soil depth, and upper- and lower-layer thicknesses for plots ($n = 3$) at each study site, with error bars (1.5 × *SD*) denoting variation among plots. Boxplot (blue) of 2.5-year continuous water table depths (1.5 × IQR) at each site located to the right of each site's soil profile

TABLE 1 Sample size (*n*), mean, and *SD* of soil physical and hydraulic properties for upper and lower soil layers, along with *p*-values from pairwise comparisons

	Upper layer		Lower layer		
	n	Mean (*SD*)	*n*	Mean (*SD*)	*p*-value
Bulk density, ρ_b (g cm^{-1})	44	0.17 (0.04)	25	0.11 (0.03)	<0.01
Organic matter (%)	44	89.65 (8.68)	25	93.28 (9.86)	<0.01
Fibre content (%)	38	11.32 (7.57)	20	16.66 (5.30)	<0.01
Total porosity, ϕ (cm^3 cm^{-3})	27	0.83 (0.07)	16	0.93 (0.06)	<0.01
Microporosity, ϕ_{mi} (cm^3 cm^{-3})	27	0.58 (0.14)	16	0.77 (0.09)	<0.01
Macroporosity, ϕ_M (cm^3 cm^{-3})	27	0.25 (0.10)	16	0.16 (0.05)	<0.01
Capillary length, λ (cm)	27	1.66 (2.34)	16	4.99 (2.65)	<0.01
θ_{1bar} (cm^3 cm^{-3})	27	0.41 (0.10)	16	0.50 (0.09)	<0.01

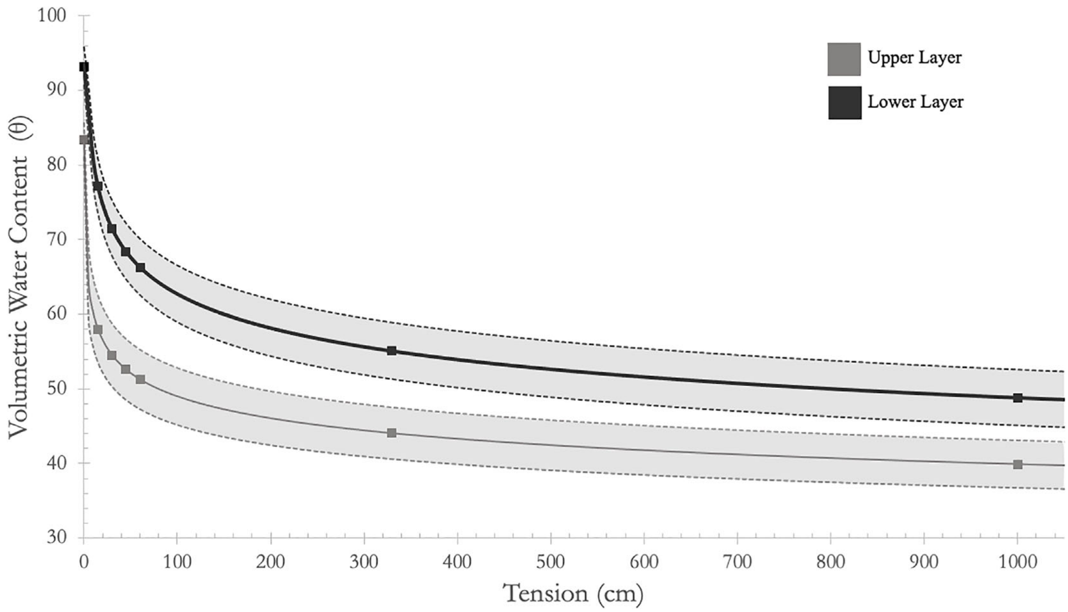

FIGURE 4 Modelled moisture release curves for upper and lower layers using mean volumetric water contents (θ) at each measured tension from all sites along with 90% confidence intervals (dashed lines)

upper layer thickness and several water table depth metrics: 25th percentile (Spearman's $r = 0.43$), mean (0.44), median (0.46) and 75th percentile (0.41). We note, however, that upper layer thickness was not significantly related to water table variation (via *SD*). Using water table depth CDFs, we also quantified the water table depth percentile that coincided with the bottom boundary of the upper layer at each plot (example in Figure 2). However, we found that 26 plots (out of a total of 45) never experienced a water table below the bottom boundary of the upper layer over the 2.5-year record (see Figure 3), precluding such an analysis for these plots where the upper layer experienced water table levels (and thus saturation) consistently within its profile. For the remaining 19 plots, the upper layer boundary corresponded to relatively deep-water table conditions; the mean boundary depth occurred at the 86th percentile of water table depth, and the median depth occurred at the 94th percentile. Only three plots had a boundary depth that corresponded to water table depths shallower than the 80th percentile. Together, these results indicate that upper layer soils at all plots experienced frequent saturation over the 2.5-year record. For example, an 86% percentile indicates that the water table was higher than the upper layer boundary 86% of the 2.5-year record and thus that some of the upper layer experienced saturated conditions 86% of the time. We repeated this analysis for nine sites (27 plots) with available water table data for 4.75 years and found similar results. For these plots, 12 never experienced water tables below the bottom boundary of the upper layer, compared with 17 when assessing their 2.5-year record. For those that did, the upper layer bottom boundary occurred on average at the 90th percentile of water table depth, thus extending the general finding to longer time periods.

While most plots frequently experienced water table levels (and thus saturation) within upper layers, sampling depth locations (1–3 per plot) for upper soil properties varied in the percentage of time saturated over the 2.5-year record (mean = 71.22%, $SD = 35.26\%$). Soil properties from up 12 per layer sampling depths—excluding bulk density, organic matter and total porosity—were significantly correlated with the percentage of time that individual profile locations were saturated (Table 4). Notably, for samples that experienced permanent

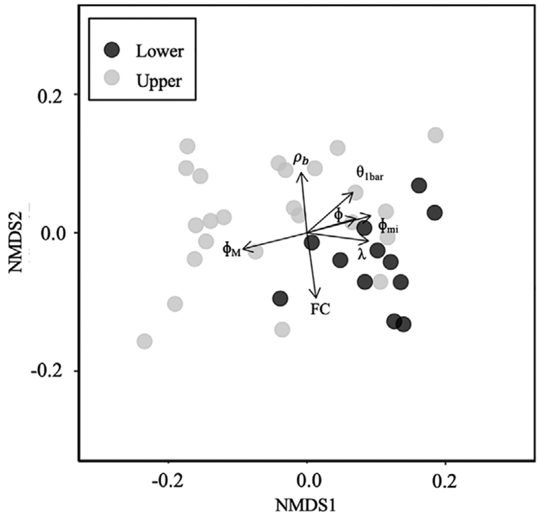

FIGURE 5 Nonmetric multidimensional scaling (NMDS) analysis with two dimensions to visualize soil property parameters with vectors present in the axes. Arrows show the direction of increasing parameter gradient, and lengths are proportional to the correlation between the parameter and ordination. Stress = 0.06. ρ_b = bulk density, θ_{1bar} = water content at 1020 cm pressure, ϕ = total porosity, ϕ_{mi} = microporosity, ϕ_M = macroporosity, λ = capillary length, FC = fibre content

TABLE 2 Factor averages of soil properties in the NMDS axes

	Axis 1	Axis 2	R^2	p-value
Bulk density (ρ_b)	−0.10	0.99	0.77	<0.001
Organic matter (OM)	−0.35	−0.93	0.14	0.09
Fibre content (FC)	0.14	−0.99	0.91	<0.001
Total porosity (ϕ)	0.96	0.28	0.54	<0.001
Microporosity (ϕ_{mi})	0.97	0.26	0.91	<0.001
Macroporosity (ϕ_M)	−0.97	−0.25	0.91	<0.001
Capillary length (λ)	0.99	−0.13	0.81	<0.001
θ_{1bar}	0.75	0.66	0.78	<0.001

Note: Columns Axis 1 and Axis 2 show correlation coefficients of soil properties with each axis and give direction to the vectors. R^2 values quantify the degree to which the ordination space affects the variable distribution, and is proportional to vector length. *p*-values show significance of soil properties on ordination of observations (points).

saturation, macroporosity approached the range of values for lower layer samples (Figure 6).

4 | DISCUSSION

Peatland drainage can increase the frequency (and depth) of aerobic conditions and also cause soil shrinkage, with associated influences to soil properties such as bulk density, pore structures and water retention capacity (e.g., Verry et al., 2011; Gebhardt et al., 2009; Schwärzel et al., 2002). Our findings demonstrate altered soil structure and lower water retention in upper layers at GDS, and that upper layer thickness is largely a consequence of past drainage as opposed to contemporary water level regimes. Notably, however, our results linking upper layer properties and percent time saturated also suggest the potential for recovery in soil properties following re-wetting efforts. Together, our findings will help inform restoration and water level management at GDS and our understanding of drained peatland soil-water interactions more broadly.

4.1 | Differences in soil physical and hydraulic properties

We found significant differences in physical properties (i.e., bulk density, organic matter, and fibre content) between upper and lower soil

TABLE 4 Spearman's correlation and corresponding *p*-values for upper layer soil properties versus percentage of time saturated

	Spearman's r	*p*-value
Bulk density (ρ_b)	−0.23	0.224
Organic matter (OM)	−0.24	0.183
Fibre content (FC)	0.42	0.016
Total porosity (ϕ)	0.24	0.266
Microporosity (ϕ_{mi})	0.60	0.002
Macroporosity (ϕ_M)	−0.67	<0.001
Capillary length (λ)	0.78	<0.001
θ_{1bar}	0.45	0.032

TABLE 3 Spearman's correlation matrix of soil property variables

	ρ_b	OM	FC	ϕ	ϕ_{mi}	ϕ_M	λ	θ_{1bar}
ρ_b	1							
OM	−0.29	1						
FC	**−0.80**	0.28	1					
ϕ	−0.15	0.11	0.17	1				
ϕ_{mi}	−0.14	0.11	0.15	**0.89**	1			
ϕ_M	0.05	−0.13	−0.09	**−0.63**	**−0.89**	1		
λ	−0.26	−0.07	0.26	**0.70**	**0.89**	**0.88**	1	
θ_{1bar}	0.20	0.21	−0.18	**0.73**	**0.87**	**−0.85**	**0.65**	1

Note: Bold values indicate a *p*-value of <0.05. ρ_b = bulk density, OM = organic matter, FC = fibre content, ϕ = total porosity, ϕ_{mi} = microporosity, ϕ_M = macroporosity, λ = capillary length, θ_{1bar} = water content at 1020 cm pressure.

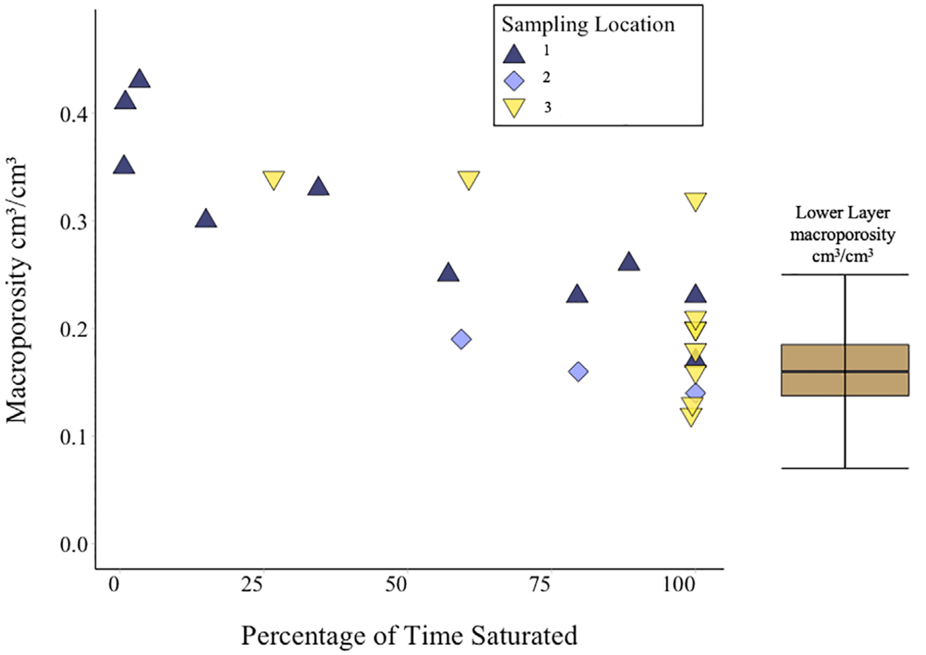

FIGURE 6 Macroporosity versus percentage of time saturated for upper layer samples at multiple depths per plot (Spearman's $r = -0.67$, p-value <0.001). Colours denote different sampling depth locations: 1 is the shallowest (e.g., 25th percentile) and 3 is the deepest (e.g., 75th percentile). Note that only one sample was collected for layers less than 40 cm thick (labelled above as sampling location 1). The distribution of macroporosity values for all lower-layer samples is also shown on the same scale

layers at GDS, likely reflecting the consequences of drainage and increased aerobic conditions. Undrained peat soils, particularly those in bog and fen systems, are typically characterized by age- and depth-dependent gradients in soil properties, where bulk density increases and organic matter and fibre contents decrease with depth (Belyea & Clymo, 2001; Boelter, 1968; Menberu et al., 2021). However, we found the opposite at GDS, with lower fibre and organic matter content and higher bulk density in the upper layer (Table 1). As bulk density is also strongly linked to the degree of decomposition (Boelter, 1968), our findings further suggest that upper layer properties have been altered and reflect increased oxidation associated with historical drainage as observed elsewhere (Kechavarzi et al., 2010; Leifeld et al., 2011).

We also found significant differences in hydraulic properties between upper and lower layers. Specifically, we observed higher macroporosity in the upper layer and thus lower moisture retention capacity. Concordantly, the upper layer had significantly lower values for metrics related to moisture retention, including capillary length, microporosity and θ_{1bar} (Table 1). It is notable, however, that our findings contrast those of studies in more northern bogs and fens (e.g., Scholtzhauer & Price, 1999; Kennedy & Price, 2005), which document similar increases in bulk density in drained soils but with *lower* macroporosities (and thus higher water retention) when compared to undisturbed peats in the same position within the soil profile. GDS soil types are characterized as sapric as opposed to fibric (more common to bog and fen systems), and sapric materials are more prone shrinkage during periods of drying (Oleszczuk et al., 2003). The shrinkage process typically causes micropores to compress in size and can lead to the creation of larger void spaces, which would explain the observed shift from massive to subangular blocky structure in the upper peats (Soil Survey Staff, 2020) as well as our opposite finding of *higher* macroporosity in disturbed soil layers. Further, we compared upper versus lower layers in our extensively drained system where undisturbed upper layers are presumed not to exist, yet other depth-dependent processes (e.g., botanical origin) may also result in depth-varying hydraulic properties (Boelter, 1968; Clymo, 1984). Had we been able to compare disturbed upper layers to undisturbed upper layers then we may have seen similar results to Price & Scholtzhauer (1999) or Kennedy & Price (2005). Future research is thus needed to compare upper layer properties between disturbed and undisturbed profiles in temperate and lower latitude peatlands and importantly those with differing (undisturbed) soil structures compared to bog and fen systems. Nonetheless, the consistently higher macroporosities and associated lower water retention in upper layers at GDS likely decrease contemporary soil moisture values above the water table, with associated consequences for carbon sequestration, plant communities and vulnerability to smouldering fire.

Increased decomposition and associated changes in physical properties can alter hydraulic properties (Verry et al., 2011; Boelter, 1968), but we suggest that drainage-induced soil shrinkage and thus changes in pore structures is an additional (if not primary) driver of upper layer hydraulic properties at GDS. First, we found no correlations between physical and hydraulic properties (Table 3); this lack of correlation also contrasts studies in disturbed northern bogs and fens (Price & Scholtzhauer, 1999; Kennedy & Price, 2005; Grover & Baldock, 2013; Liu et al., 2020), which have consistently observed evident relationships physical properties (particularly bulk density) and hydraulic properties, including water retention and hydraulic conductivity (not measured here). Further, NMDS analysis revealed that hydraulic properties were largely described by Axis 1, which generally grouped upper- and lower-layer samples with increasing water retention properties opposing macroporosity (Figure 5, Table 2). Together, these results indicate that altered pore structure has the strongest effect on water retention, and suggest that such may be a consequence of drying-induced shrinkage as observed by others (Gebhardt et al., 2012; Peng et al., 2007; Peng & Horn, 2007; Schwärzel et al., 2002). While we did not directly measure shrinkage, we observed some evidence of it by assessing changes in saturated mass following each tension when developing moisture release curves. We found that saturated mass decreased in lower layer samples by ~11% (range = 3% to 23%) over the entire range of tensions compared to ~6% (range = 0% to 14%) in upper layer samples. Such declines in saturated mass during drying may indicate volume changes from shrinkage, and larger decreases in lower layer samples suggest that the upper layer had already experienced more shrinkage prior to our analysis. These findings are further supported by similar work (Schwärzel et al., 2002; Wallor et al., 2018), where greater drying-induced shrinkage was observed in undisturbed peats than in drained peats. Still, continued research is needed to directly observe the potential for drainage-induced shrinkage in GDS and other peatland types, as well to further assess how different organic soils respond in regards to pore structure (increased versus decreased macroporosity) and associated water retention properties.

4.2 | Relating upper soil layer properties to contemporary water tables

Portions of GDS had been ditched and drained for ca. 200 years before the U.S. Fish and Wildlife Service began installing water control structures in the mid-1980s to raise the water table and mitigate the impacts of drainage. Yet, the extent to which upper layer thicknesses and properties respond to these contemporary water levels remains largely unknown. We explored relationships between current water table regimes and upper soil layer thickness across 45 sampling points (15 sites, each with 3 plots) and over a 2.5-year water table record (Figure 1). We found some significant, albeit weak to moderate, relationships between upper layer thickness and several water table metrics. Across all plots, however, the bottom boundary of the upper layer was much lower than the mean water table. Further, across plots, the majority of water table levels occurred above the bottom boundary of the upper layer, indicating frequent saturation within the upper layer. Of our 45 plots, daily water tables only fell below the upper layer at 19 plots over the 2.5-year observation period. For these 19 plots, the upper layer boundary corresponded with extremely deep-water table conditions (water table probability = 86th percentile). As water table position influences peat decomposition (particularly in bog systems; Ingram, 1978; Clymo, 1984) and the degree of potential shrinkage (Peng et al., 2007; Peng & Horn, 2007), there is an expected relationship between upper layer depth and water table depth (Whittington & Price, 2006). Thus, the observation that upper peats at GDS are typically deeper than most recently observed water tables suggests upper layer thickness is an artefact of historical drainage as opposed to contemporary water table depths.

While our results suggest that upper layers developed during periods of historical drainage, there is some evidence of potential recovery with current and higher water level regimes. Other studies have demonstrated limited recovery following drying-induced shrinkage (Gebhardt et al., 2009; Gebhardt et al., 2012; Peng et al., 2007; Peng & Horn, 2007), yet our observed associations between some soil properties and saturation frequency may suggest otherwise for upper layers at GDS. Taking advantage of multiple sampling depths within upper layers, we observed significantly lower macroporosity and higher microporosity in depth locations with greater saturation frequency, along with concordant increases in key water retention parameters (Table 4). Notably, macroporosity values in upper layer samples experiencing near-constant saturation approached observed values for lower layer samples. Thus, these results indicate potential recovery to pre-disturbed conditions and mostly likely due to recovery from shrinkage rather than from increased oxidation, which is largely irreversible. This potential recovery of degraded peat soils remains a key research need.

4.3 | Implications for management and future study

At GDS and other drained peatlands, water control structures are often used to restore historical water levels and achieve soil moisture regimes necessary for restoring vegetation communities, reducing soil oxidation, and decreasing fire vulnerability (Chimner et al., 2016; Wurster et al., 2016). Our findings indicate that the upper soil layer at GDS has reduced water retention and thus greater vulnerability to drying, even under relatively low tensions (and thus shallower water table depths). Consequently, maintaining even higher water levels (e.g., at or above soil surface) may be needed to overcome lower water retention capacity and to achieve soil moisture goals. At the same time, higher water levels may also potentially drive long-term recovery of water retention properties, yet continued study is needed to understand the timescale and extent of this potential recovery. Further, our findings of decreased water retention in drained sapric soils are in contrast to observations in disturbed bog and fen where more fibric soils occur. As such, there is a need for continued work across the diversity of peatland systems to understand how differences in peat characteristics can influence their response to drainage and subsequent restoration.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to report regarding the contents of this research and subsequent manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

ASTM International. (2008). *ASTM D4531-86: Standard test methods for bulk density of peat and peat products*. ASTM International.

ASTM International. (2013). *ASTM D1997-13: Standard test method for laboratory determination of fiber content of peat samples by dry mass*. ASTM International.

ASTM International. (2016). *ASTM D6836-16: Standard test methods for determination of the soil water characteristic curve for desorption using hanging column, pressure extractor, chilled mirror hygrometer, or centrifuge*. ASTM International.

ASTM International. (2020). *ASTM D2974-20: Standard test methods for determining the water (moisture) content, ash content, and organic material of peat and other organic soils*. ASTM International.

Atkinson, R. B., DeBerry, J. W., Loomis, D. T., Crawford, E. R., Belcher, R. T., Brown, D. A., & Perry, J. E. (2003). Water tables in Atlantic white cedar swamps: Implications for restoration. In *Atlantic white cedar restoration ecology and management, proceedings of a symposium* (pp. 137–150). Christopher Newport University, Newport News.

Belyea, L. R., & Clymo, R. S. (2001). Feedback control of the rate of peat formation. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, *268*(1473), 1315–1321. https://doi.org/10.1098/rspb.2001.1665

Boelter, D. H. (1968). Important physical properties of peat material. In *Proceedings of the Third International Peat Congress* (pp. 150–156). Ottawa, Canada: Department of Energy, Mines and Resources.

Brady, N. C., & Weil, R. R. (2010). *Elements of the nature and properties of soils* (3rd ed.). Pearson Educational International.

Brooks, R. H., & Corey, A. T. (1964). Hydraulic properties of porous media and their relation to drainage design. *Transactions of ASAE*, *7*(1), 26–28. https://doi.org/10.13031/2013.40684

Chimner, R. A., Cooper, D. J., Wurster, F. C., & Rochefort, L. (2016). An overview of peatland restoration in North America: Where are we after 25 years? *Restoration Ecology*, *25*(2), 283–292. https://doi.org/10.1111/rec.12434

Clymo, R. S. (1984). The limits to peat bog growth. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, *303*(1117), 605–654. https://doi.org/10.1098/rstb.1984.0002

Clymo, R. S., Turunen, J., & Tolonen, K. (1998). Carbon accumulation in peatland. *Oikos*, *81*(2), 368–388. https://doi.org/10.2307/3547057

Dane, J. H., & Hopmans, J. W. (2002). Soil solution phase: Laboratory. In *Methods of soil analysis part - 4 physical methods: SSSA book series - 5* (pp. 675–720). Soil Science Society of America.

Frandsen, W. H. (1997). Ignition probability of organic soils. *Canadian Journal of Forest Research*, *27*(9), 1471–1477. https://doi.org/10.1139/x97-106

Gauthier, T. L., McCarter, C. P., & Price, J. S. (2018). The effect of compression on sphagnum hydrophysical properties: Implications for increasing hydrological connectivity in restored cutover peatlands. *Ecohydrology*, *11*(8), 1–11. https://doi.org/10.1002/eco.2020

Gebhardt, S., Fleige, H., & Horn, R. (2009). Shrinkage processes of a drained riparian peatland with subsidence morphology. *Journal of Soils and Sediments*, *10*(3), 484–493. https://doi.org/10.1007/s11368-009-0130-9

Gebhardt, S., Fleige, H., & Horn, R. (2012). Anisotropic shrinkage of mineral and organic soils and its impact on soil hydraulic properties. *Soil and Tillage Research*, *125*, 96–104. https://doi.org/10.1016/j.still.2012.06.017

Grover, S. P. P., & Baldock, J. A. (2013). The link between peat hydrology and decomposition: Beyond Von Post. *Journal of Hydrology*, *479*, 130–138. https://doi.org/10.1016/j.jhydrol.2012.11.049

Holden, J., Evans, M. G., Burt, T. P., & Horton, M. (2006). Impact of land drainage on peatland hydrology. *Journal of Environmental Quality*, *35*(5), 1764–1778. https://doi.org/10.2134/jeq2005.0477

Ingram, H. A. P. (1978). Soil layers in mires: Function and terminology. *Journal of Soil Science*, *29*(2), 224–227. https://doi.org/10.1111/j.1365-2389.1978.tb02053.x

Ingram, R. L., & Otte, L. J. (1981). Peat deposits of dismal swamp pocosins: Camden, Currituck, Gates, Pasquotank, and Perquimans counties, North Carolina. In *Technical report for U.S. Department of Energy and North Carolina Energy Institute*.

Joosten, H., & Clarke, D. (2002). *Wise use of mires and peatlands: Background and principles including a framework for decision-making*. International Mire Conservation Group.

Kechavarzi, C., Dawson, Q., & Leeds-Harrison, P. (2010). Physical properties of low-lying agricultural peat soils in England. *Geoderma*, *154*(3–4), 196–202. https://doi.org/10.1016/j.geoderma.2009.08.018

Kennedy, G. W., & Price, J. S. (2005). A conceptual model of volume-change controls on the hydrology of cutover peats. *Journal of Hydrology*, *302*(1-4), 13–27. https://doi.org/10.1016/j.jhydrol.2004.06.024

Klute, A. (1986). Water retention: Laboratory methods. In *SSSA book series methods of soil analysis: Part 1—Physical and mineralogical methods*. Madison, WI: Soil Science Society of America. https://doi.org/10.2136/sssabookser5.1.2ed.c26

Lappalainen, E. (1996). General review on world peatland resources. In *Global peat resources* (pp. 53–56). International Peat Society.

Leamer, R. W., & Shaw, B. (1941). A simple apparatus for measuring non-capillary porosity on an extensive scale. *Agronomy Journal*, *33*(11), 1003.

Leifeld, J., & Menichetti, L. (2018). The underappreciated potential of peatlands in global climate change mitigation strategies. *Nature Communications*, *9*, 1071. https://doi.org/10.1038/s41467-018-03406-6

Leifeld, J., Müller, M., & Fuhrer, J. (2011). Peatland subsidence and carbon loss from drained temperate fens. *Soil Use and Management*, *27*(2), 170–176. https://doi.org/10.1111/j.1475-2743.2011.00327.x

Levy, G. F. (1991). The vegetation of the great dismal swamp: A review and an overview. *Virginia Journal of Science*, *42*(4), 411–418.

Lewis, I. F., & Cocke, E. C. (1929). Pollen analysis of the dismal swamp peat. *Journal of the Elisha Mitchell Scientific Society*, *45*, 37–58.

Liu, H., Price, J., Rezanezhad, F., & Lennartz, B. (2020). Centennial-scale shifts in hydrophysical properties of peat induced by drainage. *Water Resources Research*, *56*(10), e2020WR027538. https://doi.org/10.1029/2020wr027538

Luxmoore, R. J. (1981). Micro-, meso-, and macroporosity of soil. *Soil Science Society of America Journal*, *45*(3), 671–672. https://doi.org/10.2136/sssaj1981.03615995004500030051x

McCarter, C. P., & Price, J. S. (2015). The hydrology of the bois-des-Bel peatland restoration: Hydrophysical properties limiting connectivity between regenerated sphagnum and remnant vacuum harvested peat deposit. *Ecohydrology*, *8*(2), 173–187. https://doi.org/10.1002/eco.1498

Menberu, M. W., Marttila, H., Ronkanen, A.-K., Haghighi, A. T., & Kløve, B. (2021). Hydraulic and physical properties of managed and intact peatlands: Application of the van Genuchten-Mualem models to peat soils. *Water Resources Research*, *57*, e2020WR028624. https://doi.org/10.1029/2020WR028624

Oksanen, J, F. G. Blanchet, M. Friendly, R. Kindt, P. Legendre, D. McGlinn, P. Minchin. (2019). Vegan: Community Ecology Package (version 2.5-6).

Oleszczuk, R., Bohne, K., Szatylowicz, J., Brandyk, T., & Gnatowski, T. (2003). Influence of load on shrinkage behavior of peat soils. *Journal of Plant Nutrition and Soil Science*, *166*(2), 220–224.

Osborn, C.C. (1919). Peat in the Dismal Swamp, Virginia and North Carolina. Contribution to Economic Geology.

Peng, X., & Horn, R. (2007). Anisotropic shrinkage and swelling of some organic and inorganic soils. *European Journal of Soil Science*, *58*(1), 98–107. https://doi.org/10.1111/j.1365-2389.2006.00808.x

Peng, X., Horn, R., & Smucker, A. (2007). Pore shrinkage dependency of inorganic and organic soils on wetting and drying cycles. *Soil Science Society of America Journal*, *71*(4), 1095–1104. https://doi.org/10.2136/sssaj2006.0156

Price, J. S., & Schlotzhauer, S. M. (1999). Importance of shrinkage and compression in determining water storage changes in peat: The case of a mined peatland. *Hydrological Processes*, *13*(16), 2591–2601. https://doi.org/10.1002/(sici)1099-1085(199911)13:16<2591::aid-hyp933>3.0.co;2-e

PRISM Climate Group (2019), Oregon State University, http://prism.oregonstate.edu.

R Core Team. (2016). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.

Reddy, A. D., Hawbaker, T. J., Wurster, F., Zhu, Z., Ward, S., Newcomb, D., & Murray, R. (2015). Quantifying soil carbon loss and uncertainty from a peatland wildfire using multi-temporal LIDAR. *Remote Sensing of Environment*, *170*, 306–316. https://doi.org/10.1016/j.rse.2015.09.017

Rein, G., Cleaver, N., Ashton, C., Pironi, P., & Torero, J. L. (2008). The severity of smouldering peat fires and damage to the forest soil. *Catena*, *74*(3), 304–309. https://doi.org/10.1016/j.catena.2008.05.008

Rydin, H., Jeglum, J., & Bennett, K. (2013). *The biology of peatlands*. Oxford University Press.

Schwärzel, K., Renger, M., Sauerbrey, R., & Wessolek, G. (2002). Soil physical characteristics of peat soils. *Journal of Plant Nutrition and Soil Science*, *165*(4), 479–486. https://doi.org/10.1002/1522-2624(200208)165:4<479::aid-jpln479>3.0.co;2-8

Sleeter, R., Sleeter, B. M., Williams, B., Hogan, D., Hawbaker, T., & Zhu, Z. (2017). A carbon balance model for the great dismal swamp ecosystem. *Carbon Balance and Management*, *12*(2), 1–20. https://doi.org/10.1186/s13021-017-0070-4

Soil Survey Staff. (2014). *Keys to soil taxonomy* (12th ed.). USDA-Natural Resources Conservation Service.

Soil Survey Staff. (2020). Natural resource conservation service, United States Department of Agriculture. *Soil Series Classification Database*.

Stewart, R. D., & Najm, M. R. A. (2018). A comprehensive model for single ring infiltration I: Initial water content and soil hydraulic properties. *Soil Science Society of America Journal*, *82*(3), 548–557. https://doi.org/10.2136/sssaj2017.09.0313

Taylor, N., & Price, J. S. (2015). Soil water dynamics and hydrophysical properties of regenerating sphagnum layers in a cutover peatland. *Hydrological Processes*, *29*(18), 3878–3892. https://doi.org/10.10002/hyp.10561

U.S. Geological Survey. (2016). National Water Information System, data available on World Wide Web (USGS Water Data for the Nation) https://waterdata.usgs.gov/nwis/gw

Verry, E. S., Boelter, D. H., Päivänen, J., Nichols, D. S., Malterer, T., & Gafni, A. (2011). Physical properties of organic soils. In *Peatland biogeochemistry and watershed hydrology at the Marcell experimental Forest* (1st ed., pp. 135–176). CRC Press. https://doi.org/10.1201/b10708

Waddington, J. M., Morris, P. J., Kettridge, G., Granath, G., Thompson, D., & Moore, P. (2015). Hydrological feedbacks in northern peatlands. *Ecohydrology*, *8*, 113–127. https://doi.org/10.1002/eco.1493

Wallor, E., Rosskopf, N., & Zeitz, J. (2018). Hydraulic properties of drained and cultivated fen soils part I - horizon-based evaluation of van Genuchten parameters considering the state of moorsh-forming process. *Geoderma*, *313*, 69–81. https://doi.org/10.1016/j.geoderma.2017.10.026

Whitehead, D. R. (1972). Developmental and environmental history of the dismal swamp. *Ecological Monographs*, *42*(3), 301–315. https://doi.org/10.2307/1942212

Whittington, P. N., & Price, J. S. (2006). The effects of water table drawdown (as a surrogate for climate change) on the hydrology of a fen peatland, Canada. *Hydrological Processes*, *20*(17), 3589–3600. https://doi.org/10.1002/hyp.6376

Wurster, F. C., Ward, S., & Pickens, C. (2016). *Forested peatland management in Southeast Virginia and Northeast North Carolina, USA, proceedings of the 15th international peat congress*. Sarawak.

Xu, J., Morris, P. J., Liu, J., & Holden, J. (2018). PEATMAP: Refining estimates of global peatland distribution based on a meta-analysis. *Catena*, *160*, 134–140. https://doi.org/10.1016/j.catena.2017.09.010

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